

Large, Complex, and Realistic Safety Clothing and Helmet Detection: Dataset and Method

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ABSTRACT

Detecting safety clothing and helmet is paramount for ensuring the safety of construction workers. However, the scarcity of high-quality datasets has impeded the development of deep learning models in this domain. In this study, we construct a large, complex, and realistic safety clothing and helmet detection (SFCHD) dataset. SFCHD is derived from two authentic chemical plants, comprising 12,373 images, 7 categories, and 50,552 annotations. We partition the SFCHD dataset into training and testing sets with a ratio of 4:1 and validate its utility by applying several classic object detection algorithms. Furthermore, drawing inspiration from the attention mechanism, we design a spatial and channel attention-based low-light enhancement (SCALE) module. SCALE is a plug-and-play component with a high degree of flexibility. Extensive evaluations of the SCALE module on both the low-light ExDark and contributed SFCHD datasets have empirically demonstrated its efficacy in enhancing the performance of detectors under low-light conditions. The dataset and code are publicly available at <https://github.com/lijfrank-open/SFCHD-SCALE>.

CCS CONCEPTS

• Computing methodologies → Object detection.

KEYWORDS

Computer Industry, Object Detection, Image Processing, Construction Safety

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1 INTRODUCTION

Safety issues are an enduring and significant concern across all industries, particularly in high-risk construction sites such as chemical plants [7]. Protective equipment, including safety clothing and helmet, is crucial for safeguarding workers in high-risk construction sites. Consequently, monitoring the usage of safety clothing and helmet in factories or construction sites is of immense safety significance and broad application value. Currently, the conventional automatic monitoring technology for safety clothing and helmet compliance is a category of sensor-based methods [6, 18]. These techniques monitor whether workers are wearing safety attire or helmets by installing sensors on them and analyzing the emitted signals. However, sensor-based approaches necessitate substantial investment in procurement, installation, and maintenance, leading to relatively high costs. In recent years, deep learning methods have gained substantial attention in computer vision [14, 26] due to their ability to self-learn useful features from large-scale, annotated training data. In particular, convolutional neural networks (CNNs) [10] are widely used for image classification and object detection. For instance, Redmon et al. [19] formulated object detection as a regression task, targeting the spatial localization of bounding boxes along with their associated class probabilities; Kolar et al. [8] applied CNNs to detect safety guardrails in construction sites; Nath et al. [15, 17] adopted CNNs to identify common construction-related objects, e.g., buildings, equipment, and workers. These endeavors have provided new insights into the monitoring of safety clothing and helmet usage.

The tremendous success of deep learning techniques is largely attributed to the availability of large-scale, high-quality, and meticulously annotated datasets. To the best of our knowledge, only two benchmark datasets for helmet detection have been published to date, namely Pictor-v3 [16] and SHWD [4]. These datasets primarily focus on the detection of helmets, with insufficient consideration given to safety clothing, leading to potential safety hazards in practical applications. Moreover, the image background in existing datasets is overly simplified, and the lighting conditions are too idealized, which is significantly different from real-world work environments. Training models directly on these datasets may lead



Figure 1: Image examples from existing datasets and SFCHD dataset. Here, (a)-(e), (f)-(j), and (k)-(n) are images selected from the Pictor-v3, SHWD, and SFCHD datasets, respectively.

to insufficient generalization capabilities in their practical applications. Therefore, these two datasets do not fully meet the needs of realistic applications.

To address the above limitations, in this work, we contribute a large, complex, and realistic high-quality safety clothing and helmet detection (SFCHD) dataset. Fig. 1 illustrates a comparison between the SFCHD dataset and existing datasets. It is evident that our SFCHD dataset offers more realistic work scenarios and more complex background environments. The SFCHD dataset comprises 12,373 images, covering 7 categories, with a total of 50,558 labeled instances, including labels such as Person, Safety Helmet, Safety Clothing, Other Clothing, Head, Blurred Clothing, and Blurred Head. To more closely mimic real-world work environments, two labels are dedicated to safety clothing and helmet under varying lighting conditions and shadow coverage, i.e., Blurred Clothing and Blurred Head. All images are captured from factory surveillance cameras, encompassing 40 different scenes across two chemical plants. Each instance has been manually annotated by professional inspectors, ensuring the accuracy of the annotations. It is worth noting that our SFCHD dataset not only provides a rich set of training samples but also serves as a benchmark for the evaluation of various detection tasks, such as small object detection and high-light object detection. To more closely approximate real-world factory environments, our dataset includes a multitude of images captured under extreme lighting conditions, such as overexposure or underexposure, reflecting the actualities of the real world.

When confronted with these extreme conditions, typical detection models often perform suboptimally. The reason is the presence

of substantial noise in the images, which has a detrimental effect on the expression of potential features. To address this challenge and enhance the model's robustness in practical application scenarios, we introduce a spatial and channel attention-based low-light enhancement (SCALE) module. The SCALE module consists of two parallel pathways, i.e., the spatial attention pathway (SAP) and the channel attention pathway (CAP). SAP focuses on extracting features from key regions in the image, enhancing the model's perception of spatial feature relationships. CAP, on the other hand, assesses the importance of different channels, learning channel-specific feature information, and assisting the model in extracting features such as color or texture from the image. SAP is composed of two scale-differentiated attention, achieving refined feature fusion and latent feature representation. The design of CAP is inspired by the channel attention mechanism in SENets [5] and incorporates a skip connection structure. Unlike many image enhancement models that require specific loss functions, the proposed SCALE is a stand-alone, plug-and-play component, which can be seamlessly integrated into detection models and trained efficiently using only detection loss.

The main contributions of this work are as follows: (1) We contribute a large, complex, and realistic high-quality benchmark named SFCHD. To our knowledge, SFCHD is the inaugural factory dataset that concurrently satisfies the detection of safety clothing and helmet. (2) To address the challenge of image detection under low-light conditions, we propose a plug-and-play module named SCALE that combines spatial attention and channel attention, enhancing feature perception in dark scenarios. (3) We apply several

classic object detection models on the SFCHD dataset for experimentation to verify its usability. Concurrently, we evaluate the SCALE module on both the ExDark and SFCHD datasets. The results substantiate that SCALE demonstrates significant enhancements in the detection of low-light images.

2 RELATED WORK

Object Detection Datasets: Despite the abundance of datasets [1, 2, 9, 12, 21] for object detection, there is a notable scarcity of those tailored to industrial settings, particularly for safety clothing and helmet detection, which are essential for ensuring engineering safety and factory compliance. While object detection algorithms have advanced, their application in industry is hindered by the lack of high-quality datasets. To our knowledge, only two open-source datasets exist for helmet detection: Pictor-v3 [16] and SHWD [4]. However, these datasets have limitations that impede their utility in both academic and industrial applications. The Pictor-v3 dataset, with only 1,330 annotated images, falls short of the large-scale data requirements for deep learning and suffers from an imbalanced composition due to a lack of negative samples and an overemphasis on color labels, potentially impairing the model’s ability to recognize helmets’ physical attributes. The SHWD dataset, with 3,241 images and 10,457 annotated instances, shows an increase in image count but consists mostly of web-downloaded images with simplistic backgrounds that do not reflect real-world complexities. Additionally, the dataset’s varied origins result in inconsistent image resolutions, which can impact model performance in practical applications. In this work, we introduce SFCHD, a high-quality, large-scale dataset that captures the intricacies of industrial scenarios, thereby facilitating the deployment of object detection algorithms in industrial safety. Table 1 compares our SFCHD dataset with existing safety helmet datasets.

Dataset	Year	#Category	#Sample	#Instance	Data Source
Pictor-v3	2020	6	1,330	9,208	Web-Mined & Crowd-Sourced
SHWD	2019	2	3,241	10,457	Web-Mined
SFCHD	2024	7	12,373	50,558	Chemical Plant

Table 1: Comparison between SFCHD and existing open-source datasets for safety helmets. In the SHWD dataset, only the images related to safety helmets are used. #Category, #Sample, and #Instance denote the numbers of categories, samples, and instances in the dataset, respectively.

3 PROPOSED SFCHD DATASET

Existing helmet datasets are insufficient to meet the requirements of practical scenarios. In this work, we collect real-world images from chemical plants and construct a high-quality safety clothing and helmet dataset.

3.1 Construction Principle

Realistic Source: Collecting realistic data is crucial for practical implementation and also makes the related studies more meaningful. Therefore, we collect on-site images of workers from surveillance cameras in two chemical plants based on specific collection criteria.

For each camera, we save one image every 10 frames when there are people in the scene. This data collection process is carried out continuously for one week, resulting in approximately 40,000 captured images as the initial data source. Subsequently, we perform data filtering while ensuring an even distribution of scenes and categories. The goal is to avoid significant imbalances in the quantity of different categories. Eventually, we retain 12,373 images as the final dataset. We employ a repetitive filtering approach to ensure the authenticity and uniformity of the data.

Data Privacy: While collecting and labeling images, we strictly follow the standard de-identification procedure by removing any sensitive information (such as time, plant location, etc.), thereby ensuring that the identities of the factory and workers remain confidential.

Diverse Scenes: To ensure a diverse range of scenarios in the dataset, we collect data from two factories, capturing nearly 40 different indoor and outdoor scenes. This approach ensures the dataset’s scene universality.

Professional Annotation: Our annotation team consists of six professional security inspectors and three graduate students who have undergone specialized training. Each instance in SFCHD is annotated by the six professional security inspectors. After annotation, three graduate students who are not involved in the labeling process are invited to review the annotations, ensuring that every image and every annotation in the dataset is completely accurate.

Various Categories: Our SFCHD dataset comprises five common categories observed at construction sites, including Person, Safety Helmet, Safety Clothing, Other Clothing, and Head. To ensure the realism of the dataset, we add annotations for objects that are difficult to differentiate. In other words, we annotate Blurred Clothing and Blurred Head for those affected by shadows and strong lighting. In this manner, the SFCHD dataset encompasses a total of 7 categories, as depicted in Table 2.

High-Low Light Scene Discriminate: Although deep learning-based object detection methods have exhibited impressive results on conventional datasets, detecting objects accurately in low-quality images captured under extreme light conditions remains a significant challenge. Thus, we provide researchers with samples of extreme light conditions to encourage the development of object detection in such scenarios. As illustrated in Fig. 2, we categorize the entire dataset into three groups: high-light, low-light, and normal-light images, based on the intensity of light sources around the person in the images. Images with an evenly distributed light source are classified as normal-light. On the other hand, images with apparent strong light sources or shadows are classified as high-light and low-light, respectively. The classification of all scenes is carried out by the three graduate students. Given the potential for different perspectives on classifying images, a voting system is used to resolve any disagreements, with the majority opinion determining the final classification outcome.

3.2 Dataset Detail

Instance Distribution: The SFCHD dataset comprises 12,373 images, 7 categories, and 50,552 annotations. The instance proportion of each category is shown in Fig. 3. As depicted in Fig. 3, the most

Category	Person	Safety Helmet	Safety Clothing	Other Clothing	Head	Blurred Clothing	Blurred Head	Total
Training	13,528	11,378	11,781	626	961	1,053	896	40,223
Testing	3,482	2,920	3,032	154	239	271	238	10,336
Total	17,010	14,298	14,813	780	1,200	1,324	1,134	50,559

Table 2: Statistics of instance distribution per category in the SFCHD dataset.

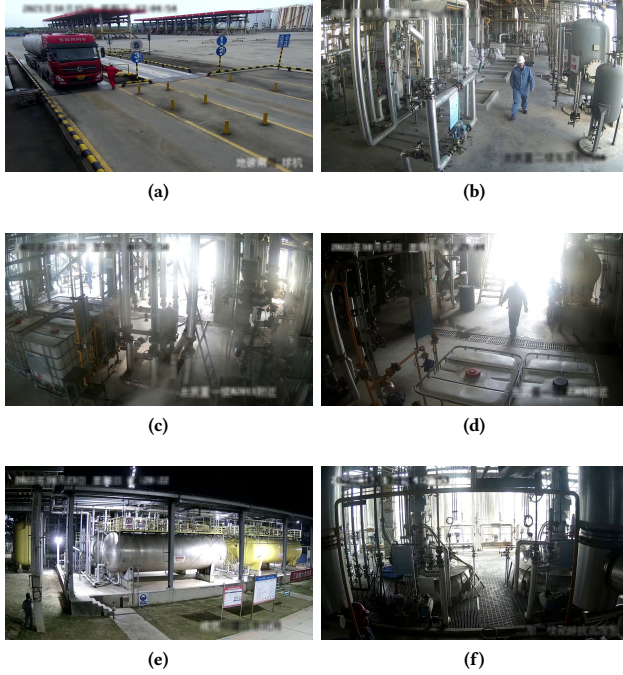


Figure 2: Image examples with different lighting conditions. Here, (a)-(b), (c)-(d), and (e)-(f) are the images based on normal-light, high-light, and low-light, respectively. In (c)-(d), it is extremely difficult to detect whether the worker is wearing safety clothing and helmet because the surrounding light is too high. Similarly, in (e)-(f), too low light around the worker makes it difficult to conduct object detection tasks.

prominent categories are Person, Safety Helmet, and Safety Clothing, accounting for about 33.64%, 28.28% and 29.30% of the dataset, respectively. Due to the rarity of violations in real-world chemical plants, situations where safety clothing or helmets are not worn are infrequent. As a result, the instances classified as Other Clothing and Head represent only a small portion of the dataset, 1.54% and 2.37%, respectively. The remaining categories, Blurred Clothing and Blurred Head, which represent instances captured in special environments, are relatively limited in quantity, accounting for 2.24% and 2.62%, respectively.

Data Partitioning: Our dataset is divided into training set and testing set, and the ratio is 4:1. We have 12,373 images in SFCHD dataset, including 9,898 images in the training set, 2,475 images in the testing set. The statistics of category distribution of training

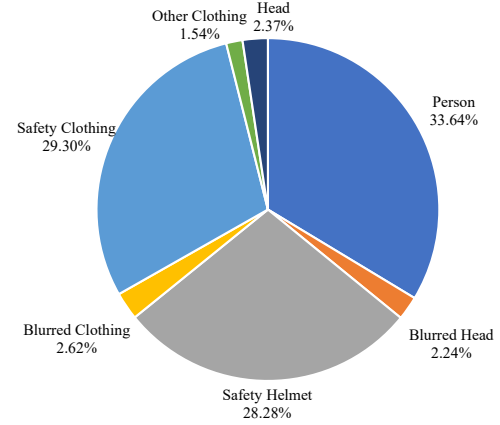


Figure 3: Proportion of instances per category in the SFCHD dataset.

and testing set under different lighting conditions are shown in Table 2.

Data Quality: To facilitate model training, all images are stored in JPG format; the average resolution of the images is predominantly 1280 * 704, with the maximum resolution reaching 1920 * 1280.

3.3 Potential Task

Small Object Detection: As per the definition provided by the international organization SPIE, a small object must not exceed 0.12% of the overall image area. Therefore, we define small objects as those whose actual bounding box occupies less than 0.1% of the entire image area, while large objects are those whose bounding box occupies more than 0.2% of the image area. The remaining objects are classified as medium-sized objects. Table 3 provides detailed statistics on the number of small objects in the dataset.

Category	Total	Large	Medium	Small
Safety Helmet	14,298	3,551	3,030	7,717
Head	1,200	95	164	941
Blurred Clothing	1,324	281	313	730
Blurred Head	1,134	15	41	1,078

Table 3: Category distribution of objects with different sizes in the SFCHD dataset.

High-Low Light Detection: A critical reason why object detection tasks are challenging under extreme light conditions is the lack of relevant datasets. According to the light conditions of the area where the object is located, we divide the images in the dataset into high-light images, low-light images, and normal-light images.

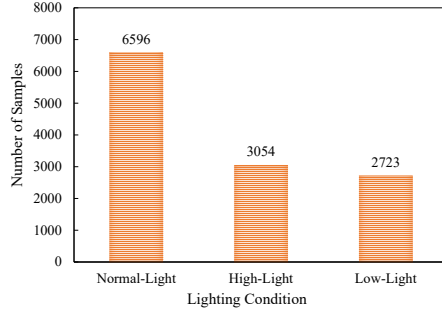


Figure 4: Quantity distribution under different lighting conditions in the SFCHD dataset.

Fig. 2 illustrates the quality of each type of image. The quantity distribution is shown in Fig. 4, and we can observe that the samples under normal-light are the most, while the samples under high- and low-light are basically the same.

4 PROPOSED SCALE MODULE

SCALE, as a plug-and-play component, integrates seamlessly into traditional object detection workflows, as depicted in Fig. 5. Specifically engineered for extreme low-light conditions, the SCALE module adaptively enhances images, consisting primarily of two components, i.e., spatial attention pathway (SAP) and channel attention pathway (CAP). The entire pipeline relies solely on detection loss for end-to-end training, simplifying the training process and enhancing efficiency. Details of the SCALE structure and related experiments can be found in the [supplementary materials](#).

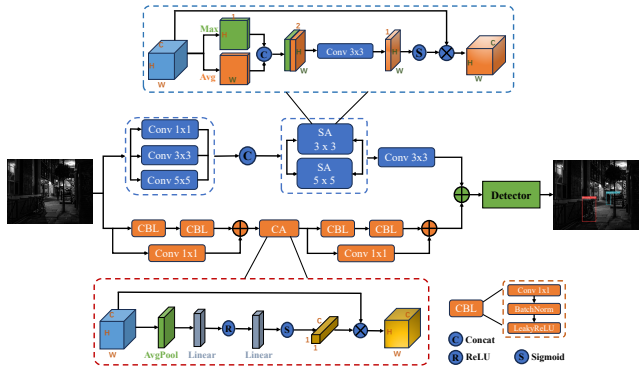


Figure 5: Network structure of spatial and channel attention-based low-light enhancement (SCALE) module.

5 EXPERIMENT

Centered around SFCHD and SCALE, our experiments are composed of three subsection. Detailed experimental setup can be found in the [supplementary materials](#).

5.1 Usability of SFCHD Dataset

In this subsection, we employ several detection models to conduct experiments on the existing helmet datasets and proposed SFCHD

dataset, substantiating the usability of our dataset. Table 4 details the performance across different categories in our dataset. YOLOv5 demonstrates superior precision in detecting the Person, Safety Helmet, and Safety Clothing categories, likely due to their rich data representation, enabling the network to capture a broader range of features. For the Other Clothing and Head categories, their distinct features from the dominant Safety categories help the network differentiate them, despite their limited proportion in the dataset. Conversely, the Blurred Clothing and Blurred Head categories, hampered by limited samples and compounded by environmental complexities like shadows and harsh lighting, pose a significant challenge for feature extraction, resulting in reduced detection accuracy.

Method	Person	Safety Helmet	Safety Clothing	Other Clothing	Head	Blurred Clothing	Blurred Head
AP(0.50:0.95)							
SSD [13]	60.2	56.5	55.7	45.2	38.4	20.5	14.2
Faster RCNN [20]	71.2	64.7	64.6	54.5	49.3	27.2	20.5
FCOS [22]	68.4	63.4	64.6	54.0	48.0	28.9	19.6
VFNet [25]	73.1	66.2	64.3	54.0	52.5	25.1	22.0
RetinaNet [11]	71.1	63.5	64.5	51.8	48.5	27.1	15.9
TOOD [3]	72.9	66.0	65.9	56.2	52.8	29.6	22.3
YOLOv5 [23]	72.7	66.4	63.7	54.9	50.7	21.2	18.9

Table 4: Performance of different categories in SFCHD.

Method	Backbone	mAP(0.50:0.95)	FPS
SSD [13]	VGG16	41.5	57.5
Faster RCNN [20]	ResNet-50	50.3	25.9
FCOS [22]	ResNet-50	49.6	27.5
VFNet [25]	ResNet-50	51.0	11.7
RetinaNet [11]	ResNet-50	48.9	27.0
TOOD [3]	ResNet-50	52.3	17.0
YOLOv5 [23]	CSPDarknet53	49.6	46.3

Table 5: Results of mAP and speed for different methods in the SFCHD dataset.

In Table 5, we compare the mAP(0.50:0.95) and inference speed of various models on the SFCHD dataset. It can be observed that TOOD achieves the highest mAP, but its FPS is only 17.0; SSD attains the highest FPS, but its mAP is lower than all other models, with a score of only 41.5%. Upon comprehensive consideration, YOLOv5 realizes the optimal balance between accuracy and speed.

Table 6 presents a comparative analysis of various methods across different datasets. It is observable that our SFCHD records the lowest scores in mAP, followed by the Pictor-V3 and SHWD datasets. This is primarily attributed to the higher complexity of our dataset, which encompasses intricate backgrounds, diverse categories, and undesirable lighting, thereby presenting greater challenges. Undoubtedly, models trained on more realistic and complex datasets, like ours, generally perform better in practical settings.

5.2 Application of SCALE on SFCHD

In this subsection, we apply the SCALE module to the SFCHD dataset to demonstrate its effectiveness in enhancing the detection quality of models on the SFCHD dataset. Table 7 outlines

Method	Backbone	Pictor-v3		SHWD		SFCHD (ours)	
		mAP(0.50)	mAP(0.50:0.95)	mAP(0.50)	mAP(0.50:0.95)	mAP(0.50)	mAP(0.50:0.95)
SSD [13]	VGG16	85.5	48.8	80.8	57.4	72.8	41.5
Faster RCNN [20]	ResNet-50	90.6	53.4	84.8	63.1	76.4	50.3
FCOS [22]	ResNet-50	89.5	52.4	85.8	63.9	76.4	49.6
VFNet [25]	ResNet-50	91.4	55.2	85.7	63.9	76.4	51.0
RetinaNet [11]	ResNet-50	90.5	54.4	85.5	63.6	75.9	48.9
TOOD [3]	ResNet-50	91.5	55.8	86.7	64.4	78.9	52.3
YOLOv5 [23]	CSPDarknet53	88.2	53.6	84.0	63.9	74.1	49.6

Table 6: Comparisons of different methods on the Pictor-v3, SHWD, and SFCHD datasets.

Method	Person	Safety Helmet	Safety Clothing	Other Clothing	Head	Blurred Clothing	Blurred Head	mAP(0.50:0.95)	mAP(0.50)
	AP(0.50:0.95)								
FCOS [22]	68.4	63.4	64.6	54.0	48.0	28.9	19.6	49.6	76.4
FCOS+SCALE	68.5	63.8	64.7	53.0	47.6	28.4	20.5	49.5	76.3
VFNet [25]	73.1	66.2	64.3	54.0	52.5	25.1	22.0	51.0	76.4
VFNet+SCALE	73.2	66.5	64.4	53.9	52.1	25.8	23.7	51.4	76.6
TOOD [3]	72.9	66.0	65.9	56.2	52.8	29.6	22.3	52.3	78.9
TOOD+SCALE	72.9	66.2	66.2	56.2	51.6	29.6	23.5	52.4	79.3
YOLOv8 [24]	73.6	64.9	67.5	58.5	45.5	32.3	23.9	52.2	77.9
YOLOv8+SCALE	74.4	66.1	68.8	58.4	47.5	33.2	25.2	53.3	78.6

Table 7: Performance improvements of the SCALE module on the SFCHD dataset.

the performance gains achieved by integrating the SCALE module into various object detection models. The results show accuracy improvements for VFNet, TOOD, and YOLOv8 when the SCALE module is applied. Specifically, VFNet with SCALE sees a 0.4% mAP(0.50:0.95) increase; TOOD with SCALE experiences a 0.4% mAP(0.50) boost; and YOLOv8 integrated with SCALE achieves a 1.1% mAP(0.50:0.95) improvement. These results underscore the SCALE module’s effectiveness in enhancing detectors’ understanding of low-light imagery. However, the SCALE module did not significantly improve FCOS model performance, possibly due to the SFCHD dataset’s extreme lighting conditions and diverse noise types, which challenge the model’s ability to enhance performance further. Thus, for complex datasets like SFCHD, the SCALE module, paired with high-performing detectors, can more effectively leverage its low-light enhancement capabilities, highlighting its potential in the object detection domain. Moreover, our contributed SFCHD dataset is expected to inspire the development of even better object detectors in the future.

6 CONCLUSION

In this study, we address the challenges of real-world construction site environments by introducing an industrial dataset, SFCHD, and a low-light image enhancement module, SCALE, to bolster artificial intelligence applications in high-risk construction scenarios. SFCHD is designed for safety clothing and helmet detection with images from actual construction sites. To validate the usability of SFCHD, we conduct experiments using various classic object detectors and compared them with other relevant datasets. Our designed SCALE is a plug-and-play module that addresses the performance limitations of existing algorithms under low-light conditions. Experimental results on both the low-light ExDarK and constructed

SFCHD datasets demonstrate that the introduction of the SCALE module has led to substantial improvements in the performance of object detectors under low-light conditions.

We intend to extend the SFCHD dataset’s application to a wider array of computer vision tasks, including instance segmentation and image classification, to broaden its utility and impact. Simultaneously, we are committed to developing innovative solutions to tackle industrial dataset challenges such as complex background noise, thereby enhancing support for industrial vision research.

ETHICS AND FAIRNESS

We ensure that the SFCHD complies with relevant privacy regulations and standards, and will not leak personal identification information or other sensitive information. All participants fully understand the purpose, methods, and potential risks of the research and participate voluntarily. During the data collection process, we have taken a variety of measures (such as different lighting conditions, different times, and different locations) to reduce the impact of bias. The constructed dataset is intended to promote the development of the industrial vision community, and it is recommended that community members use this dataset solely for scientific research purposes.

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